

Andrew Mata

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Introduction

I am a PhD candidate in Computer Science at Stony Brook University in the Reliable Systems Lab under the supervision of Dr. Stanley Bak. My research focuses on AI safety and robustness of reinforcement learning agents and other learning-enabled systems.

My work develops methods for adversarial evaluation, robustness certification, and systematic discovery of failure modes in machine learning systems. In particular, I design scalable techniques for stress-testing reinforcement learning agents operating in complex environments, including adversarial perturbation frameworks, robustness certification methods, and surrogate model-based falsification tools. These approaches aim to improve the reliability, robustness, and safety of AI systems deployed in real-world and safety-critical settings.

Research Areas

Reinforcement Learning, AI Safety, Adversarial Machine Learning, Neural Network Verification, Learning-Enabled Systems

Education

Ph.D.	Computer Science	Stony Brook University	(ongoing)
M.S.	Computer Science	Stony Brook University	2023
B.A.	Computer Science and Math	Cornell University	2020

Academic Employment

Stony Brook University , NY, USA	Fall 2020-Present
Research Assistant, 2021-Present	
Teaching Assistant, 2020	
Cornell University , NY, USA	Fall 2018-19
Teaching Assistant	

Internships

Air Force Research Laboratory (AFRL) , OH, USA	Summer 2024
Research Intern, Autonomy Capability Team (ACT3)	
Conducted research on adversarial robustness and safety evaluation for learning-enabled autonomous systems, with a focus on reinforcement learning-based controllers.	

Distinctions

SBU GAANN Fellowship, 2021

Best Artifact Evaluation Package Award at ICCPS, 2024 (for Zero-One Attack adversarial RL evaluation framework)

Selected Projects

Zero-One Attack: Adversarial Evaluation of Reinforcement Learning Agents (code)

Developed a framework for generating worst-case perturbations in both observation space and activation timing to systematically evaluate the robustness of reinforcement learning agents. Demonstrated stronger degradation than prior state-of-the-art attack baselines on MuJoCo control benchmarks. Implemented in Python/PyTorch and released as an artifact evaluation package that won the Best Artifact Evaluation Package Award at ICCPS.

Koopman Surrogate Falsification (FReaK Tool)

Designed a scalable framework for discovering failure cases in controllers using Koopman-based surrogate models combined with reachability analysis and MILP encodings of Signal Temporal Logic specifications. The method efficiently identified low-robustness trajectories and outperformed competing tools in 16 of 19 benchmarks in the 2022 ARCH falsification competition.

Symbolic Gaussian Smoothing

Developed a deterministic robustness certification method that replaces Monte Carlo randomized smoothing with symbolic moment propagation, enabling analytical computation of output distribution statistics in a single forward pass. The approach provides scalable robustness guarantees for neural network models without relying on stochastic sampling.

Adversarial Drone Flocking

Investigated adversarial observation-noise attacks on neural network-based multi-agent flocking policies. Designed a secondary-agent attacker capable of perturbing agent observations during execution and evaluated system robustness through alternating attacker-controller training.

Technical Skills

Python, PyTorch, Reinforcement Learning, Stable-Baselines3, Gymnasium, MuJoCo, NumPy, MILP Optimization, Neural Network Verification

Selected Publications

Journal Articles

1. Veena Krish, Andrew Mata, Stanley Bak, and Amir Rahmati. 2023. Provable Observation Noise Robustness for Neural Network Control Systems. *Research Directions: Cyber-Physical Systems*, vol. 1, 2023, doi:10.1017/cbp.2023.2

Conference Articles

2. Stanley Bak, Sergiy Bogomolov, Abdelrahman Hekal, Veena Krish, Andrew Mata, Amir Rahmati. 2024. Zero-One Attack: Degrading Closed-Loop Neural Network Control Systems using State-Time Perturbations. ACM/IEEE In-

ternational Conference on Cyber-Physical Systems (ICCPs) **Best Artifact Evaluation Package Award**

3. Stanley Bak, Sergiy Bogomolov, Abdelrahman Hekal, Niklas Kochdumper, Ethan Lew, Andrew Mata, Amir Rahmati. 2024. Falsification using Reachability of Surrogate Koopman Models. ACM International Conference on Hybrid Systems: Computation and Control (HSCC)
4. Andrew Mata, Ali Arjomandbigdeli, and Stanley Bak. 2025. Symbolic Gaussian Smoothing. Proceedings of the ACM/IEEE 16th International Conference on Cyber-Physical Systems (ICCPs)
5. Stanley Bak, Abdelrahman Hekal, Niklas Kochdumper, Ethan Lew, Andrew Mata, Amir Rahmati. 2024. Fast Koopman Surrogate Falsification Using Linear Relaxations and Weights. International Symposium on Automated Technology for Verification and Analysis (ATVA)
6. Ali Arjomandbigdeli, Andrew Mata, and Stanley Bak. 2024. Verification of Neural Network Control Systems in Continuous Time. 7th Symposium on AI Verification (SAIV)

Teaching Experience

Robotics/AI Methods/Reinforcement Learning	Summer 2022-Present
Teacher at Sigma Camp USA and Sigma Camp ETAPA Brazil	
Computer Science 101, 150, 201	Fall 2022-25
Teacher at SchoolNova Enrichment Program	
CSE 150: Foundations of Computer Science - Honors	Fall 2020
Teaching Assistant, with Dr. R. Sekar at Stony Brook University	
CS 4860: Applied Logic	Fall 2018
Teaching Assistant, with Dr. Robert Constable at Cornell University	

References

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Dr. Amir Rahmati, Assistant Professor, Department of Computer Science, Stony Brook University
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